

Enhancing Multimodal Interaction Between Performers and Audience Members During Live Music Performances

Kouyou Otsu
Saitama University, Saitama, Japan
otsu@cv.ics.saitama-u.ac.jp

Jinglong Yuan
Saitama University, Saitama, Japan
enkeiryuu@yahoo.com

Hisato Fukuda
Saitama University, Saitama, Japan
fukuda@cv.ics.saitama-u.ac.jp

Yoshinori Kobayashi
Saitama University, Saitama, Japan
yoshinori@hci.ics.saitama-u.ac.jp

Yoshinori Kuno
Saitama University, Saitama, Japan
kuno@cv.ics.saitama-u.ac.jp

Keiichi Yamazaki
Saitama University, Saitama, Japan
BYI06561@nifty.com

ABSTRACT

Live performance provides a good example of enthusiastic interaction between people gathered together in a large group and one or more performers. In this research, we focused on elucidating the mechanism of such enthusiastic group interaction (collective effervescence) and how technology can contribute to its enhancement. We propose a support system for co-experience and physical co-actions among participants to enhance enthusiastic interaction between performers and audiences during live performances. This system focuses on a physical synchronization between the performer and the audience as the key that generates enthusiastic interaction during a live performance. Also, it supports enhanced bidirectional communication of the performer's actions and the audience's cheering behaviors. An experiment in an actual live performance environment in which collective effervescence was already occurring demonstrated that the bidirectional communication and visualization of physical movements in the proposed system contributed to the further unification of the group.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices;; Collaborative and social computing; Collaborative and social computing devices.

KEYWORDS

interaction support system, live music performance, collective effervescence

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1 INTRODUCTION

Multiparty interaction has special value based on shared experience that cannot be replicated individually. In such multi-party interactions, people gather and interact based on their common experience, promoting unified action as a group with a shared purpose; in the field of psychological study, this is called *collective effervescence* [3]. Collective effervescence in the real world can be seen in various events involving large groups, such as political rallies and concerts. Social networking systems (SNS) such as Twitter and YouTube have become a powerful means to bring about energetic emotional connections among masses of people with specific values in the virtual environment [13, 26]. Such collective effervescence has a strong effect on motivation, especially in situations where a huge group of people has the ability to interact. Therefore, technological support to promote or guide it would open doors to many possibilities, such as contributing to the realization of valuable interactions and the prevention of conflicts within various large groups, such as at live events, political rallies, etc.

One example where collective effervescence directly promotes the enhancement of user experience can be seen in live music performance. The number of live concerts where people can interact with actual performers has increased worldwide [8, 19]. This increase indicates that many audience members think it important to feel the empathy gained by meeting a real performer and enjoying music with him/her and friends who are fans supporting the same performer. Not only on-site support but interaction through comments on live streaming, such as on YouTube Live, are avenues that contribute to collective effervescence among participants of live performances [13, 24]. However, how collective effervescence is formed online differs from how it can occur in on-site interactions and does not provide the same participatory experience as in the local environment [12, 27]. In a live performance by a pop musician, the audience dances, sings, and claps along with the performance, interacting with the performer and other audience members. Some fans also use cheering items, such as light sticks, to express their belonging and support for a group of performers. The common experience among on-site participants during a live performance and the imitation and synchronization of physical actions that provoke it enhances a sense of solidarity among participants and promotes enthusiastic interaction [16, 23]. Even for remote participants in live streaming environments, promoting the synchronization of participants with each other's actions, as with those at the site, can enhance the participation of diverse people by breaking down the

barriers of distance and making everyone feel like a part of the event.

In this study, we focus on live music performances as a good example of many people gathering and causing a collective energy and aim to realize a framework for promoting enthusiastic interaction by providing a common support system that involves “physical synchronization” for people in both remote and local environments. The key motivation of this framework is to provide an equivalent system of participation in both environments. By enhancing the experience at the site and providing the same sense of participation for those online, the framework aims to break down barriers in addition to enhancing the value of each individual’s experience.

As a technology to support live performance, some consumer products, such as Xirobands [30] and GlowMotion [17], attempt to realize integrated performance using light-emitting devices through wireless control. As mentioned earlier, cheering items, such as light sticks, can provide active engagement with the other participants. However, the automated staging by the event organizer included in these systems impairs the opportunity for the audience to participate voluntarily. Similarly, there are proposals for methods to collect input from the audience [1, 6], reflect this in the performance, and integrate the movements of a performer or the audience as CG effects [4, 5, 15, 22, 25]. However, these attempts do not focus on improving the interaction between the audience and the performer and supporting co-experience or co-action.

Many studies have been conducted in the context of telecommunication support to enhance the connection between individuals by supporting the cognition of others through sensory feedback such as haptics [2, 10, 11, 18]. Such sensory-sharing technology can be applied to support co-experience and co-activity that brings about collective effervescence, as it facilitates the recognition of the other person’s presence and experience and may also be used to provide a common experience between the local and remote environments. In the context of game entertainment, several attempts have been made to support user co-experience-based interaction by incorporating the psychological concept of *mirroring* [9, 14, 20, 21]. It has also been suggested that such a framework for inducing physically synchronized experiences can enhance intimacy in personal communication.

In this paper, we clarify whether using our interactive devices to support bidirectional communication would be able to enhance collective effervescence and induce new behaviors of participants. We first focus on collective effervescence at the local site of live performances as an initial study and we examined whether the proposed system could enhance this interaction system, which has already formed a collective energy. The field experiments in this study were conducted before the COVID-19 pandemic, and the experiments were conducted with attention to the safety of the participants.

2 LIVE PERFORMANCE SUPPORTING SYSTEM

2.1 SYSTEM OVERVIEW

Fig. 1 shows an overview of the proposed system. This system aims to improve the sense of unity and the experience value in multi-party interaction by supporting an information transfer between the

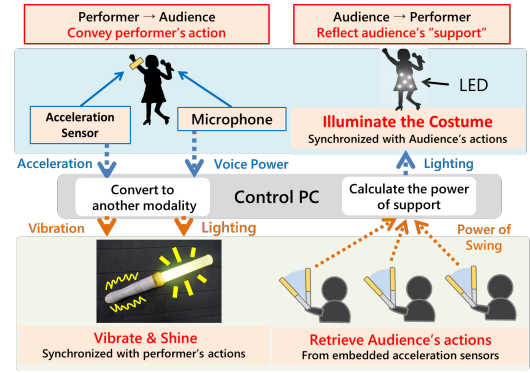


Figure 1: Overview of the proposed system.

performer and audience that is inherent in a live performance. As previously mentioned, one of the goals of this research is to provide a support framework that allows both local and remote audiences to feel a common participation experience. This study focused on the cheering item (a light stick device), which is often used by audiences in live performances. We constructed a system that enables the audience to experience a performer’s actions as vibration and light from the device; an audience member can also communicate their support more explicitly for a performer and others by physically swinging the device. In this system, the physical movement and the performer’s singing voice are acquired respectively from a wireless accelerometer and microphone attached to the performer’s costume. The information is fed back to the audience member’s device as vibration and lighting in real time. The system also obtains the swinging of the device by the audience member from the built-in accelerometer and feeds it back to the performer’s costume as a lighted element. In this paper, we have developed a prototype of the proposed system and conducted an evaluation experiment. In the following, we describe the implementation of the prototype system.

2.2 HARDWARE IMPLEMENTATION

To implement the prototype system, we developed a dedicated user device, see Fig. 2 (left), by modifying a consumer light stick (King Blade X10 III, Ruifan Ltd.) that is often used in live music performances. The user device consists of an input device (built-in accelerometer), feedback devices (RGB-full-color LED and vibration motor), and a microcontroller (TWE-Lite, Mono-Wireless Inc.) that supports Zigbee-based 2.4GHz wireless communication. The user device is in constant communication with an external control PC. The external control PC manages the built-in feedback module on user devices in real time via wireless communication. It can also acquire data from the built-in accelerometer on user devices in real time. Through this user device, the system can communicate the performer’s physical activity to the user and obtain information about the user’s physical activity.

Fig. 2 (right) shows the setting of the performer’s side of the prototype system. This consists of a wireless wristband-type acceleration sensor that captures the physical movements of the performer’s actions and an electrically lighted stage costume that feeds back the

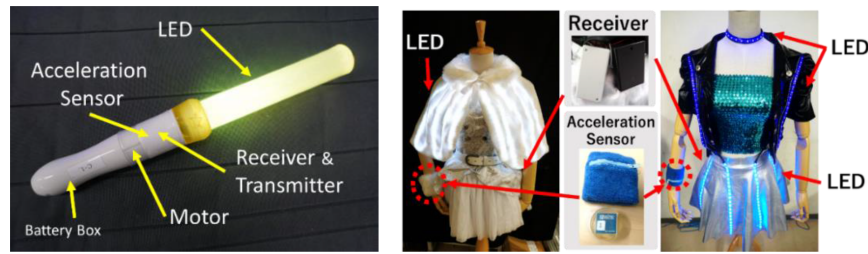


Figure 2: The user device (left) and the performer's setting (right) for the prototype system

intensity of the audience's swinging actions. A small accelerometer tag (TWE-Lite 2525A, Mono-Wireless Inc.) that supports 2.4GHz wireless communication was used as a sensor to acquire the performer's actions. Through this sensor, it is possible to acquire information on the performer's arm swing in real time at a speed of about 30 Hz. The performer's illuminated costume is composed of about 200 LEDs, and it is possible to control the color and lighting state by wireless communication in real time.

2.3 SOFTWARE IMPLEMENTATION

The activity information of the performer and the audience obtained through the devices mentioned in the previous section are transmitted wirelessly to the control PC. The control PC manages the communication between the performer and the audience. The software implemented on the PC is responsible for converting the modality of the information from the performer to the audience and aggregating the audience members' activity information. The software acquires the average value of the acceleration variations obtained from the accelerometer worn by the performer and the audio intensity of the microphone in real time and transmits control commands containing the converted vibration motor and emission intensity information to up to 100 user devices in real time. In the data conversion process, the magnitudes of the two types of information acquired from the performer are rescaled (to a range of 0–255). These rescaled values are set as the intensity of vibration and luminescence transmitted to the audience member devices.

The acceleration data transmitted from all the user devices are also sequentially acquired. The total value acquired from each audience member's acceleration data is then used as the intensity of the audience's cheering action at that moment. The intensity value is utilized as a reference value to determine the lighting intensity of the performer's costume and the number of LEDs to be lit. In this way, the light feedback to the performer's costume is provided according to the audience's physical action obtained from the audience member's devices. The feedback time between the performer's performance and the audience's support was designed to be within a maximum of one second in the two experimental environments described below.

3 EXPERIMENT UNDER A CONTROLLED LIVE PERFORMANCE SITUATION

We first experimented with a form of live music performance with some restrictions, including two types of tests to evaluate the effectiveness of vibration feedback to the audience synchronized

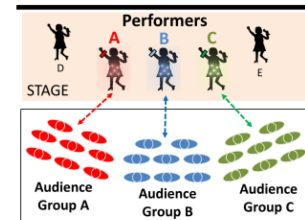


Figure 3: Experiment scene (above) and experimental setup (below)

with the performer's action and the usefulness of the interactive information transmission. The experiment was conducted on January 10, 2018, in a university hall; Fig.3 (above) shows the scene of the experiment. For the experiments, a temporary stage and audience seats were prepared, and an environment that resembles an actual live music concert was built. Five students belonging to a dance club assisted the experiment as performers. Also, 24 students participated as audience members. Of the 24 participants, 19 had previously attended a live performance, and 13 of them attended at least one a year. In the experiments, the 24 participants were divided into three groups of eight participants each. In the experiment, three of the five performers were each individually assigned to one of the groups. Therefore, the 24 participants were instructed to specifically support the performer to which they were assigned (i.e., they were "fans" of that particular entertainer). Fig. 3 (below) shows the setting of the venue and how the audience was divided based on their assigned performer to support.

3.1 Experiment 1: Survey of the validity of vibration feedback synchronized with the performer's actions

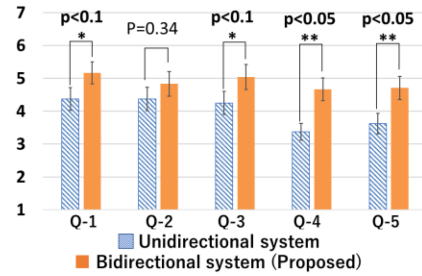
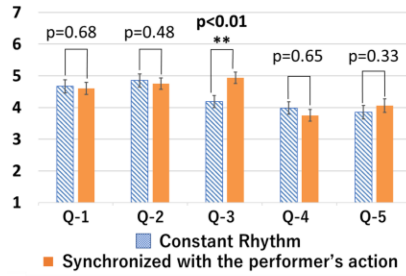
In this experiment, we evaluated the effectiveness of vibration feedback in the devices linked to the performers' movements in four

Table 1: The order of the songs and feedback patterns from Experiment 1

No.	Song	Feedback pattern
1	Song A	Synchronized with the performer's actions
2	Song B	Constant rhythm
3	Song A	Constant rhythm
4	Song B	Synchronized with the performer's actions

Table 2: Items in the questionnaire

Question No.	Question
Q-1	Did you feel more immersed by using the system?
Q-2	Were you able to concentrate on the live music performance?
Q-3	Did you feel a sense of unity with the assigned performer?
Q-4	Did you feel a sense of unity with other audience members?
Q-5	Did you feel a sense of unity with audience members in your assigned group?

**Figure 4: Mean scores of the questionnaire items and p -values of Experiment 1 (left) and Experiment 2 (right).**

separate trials (i.e., four different songs with different conditions). The order of the performances and the type of feedback are shown in Table 1. The audience members experienced two vibration patterns, one linked to the performer's performance and the other vibrating at a constant rhythm regardless of the performer's motions. They then completed a questionnaire that used a 7-point evaluation scale (1: completely disagree, 7: completely agree), as shown in Table 2. For the questionnaire to the participants, we translated the concept of collective effervescence and asked questions about enthusiasm, concentration, and sense of unity with others, which may be factors in forming it in live performance.

Because this experiment was designed to evaluate the differences in the audience's impressions due to the differences in the vibration patterns, the system was operated only for the unidirectional transmission of information from the performers to the audience (without audience feedback to the performers' illumination costumes). Fig. 4 (left) shows the mean scores and p -values derived from Mann-Whitney U tests for each item from the questionnaire. As shown in the result for Q-3, the vibration linked to the performer's body movement made a significant difference in the audience member feeling a better sense of unity with the performer.

3.2 Experiment 2. Survey of the validity of bidirectional communication support and its effectiveness

This experiment was conducted following Experiment 1. In this experiment, the same song was sung by the performer twice, and the participants experienced two different systems in each. In the first trial, the system was operated as a unidirectional system that only provided the audience with vibration and light feedback from the performers (i.e., the electric costumes were lit at a constant brightness). The second trial was operated as a bidirectional system that supported bidirectional communication between the performers and the audience in which the audience member's support altered the illumination of the costumes (proposed method). After the experiment, the questionnaire shown in Table 2 was completed again. Fig. 4 (right) shows the mean scores and the p -values derived from the Mann-Whitney U tests for each item from the questionnaire. Significant differences were observed in all the questions between the unidirectional and bidirectional uses of the system. The greatest increases were seen for Q-4 and Q-5. The results showed that the bidirectional information transmission contributed more to enhancing the connection between the performers and audience and the sense of unity among the audience than the unidirectional mode.

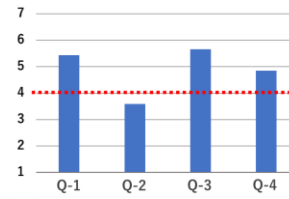


Figure 5: Experiment scene (left) and Mean scores from the questionnaire (right).

4 EVALUATION UNDER A REALISTIC LIVE PERFORMANCE SITUATION

The initial two experiments constituted a basic verification of the system with students as audience members. A more thorough investigation of the system’s effectiveness required verification with professional performers and actual fans in a realistic setting. Therefore, we evaluated the system with professional performers and their fans in a way that was faithful to an actual live music performance. The evaluation was conducted on January 16, 2019, at a university hall equipped with a stage and sound equipment. A total of 50 members of the performers’ fan club were recruited via a website as participants in the evaluation. Five of the seven performers were each individually assigned to 50 participants as in the previous experiments. The audience’s seating position was not specified, but they were informed of the performer assigned to them in advance. In this evaluation, the participants experienced two types of devices: a normal light stick (without enhancement) and our proposed system, which supported the vibration feedback synchronized with the performers’ actions and the lighting feedback to each assigned performer’s illuminated costume based on the intensity of the swinging action of the device. After the event, the participants were asked to complete the items Q-1 to Q-4 of the questionnaire shown in Table 2 from the viewpoint of “How did you feel the proposed system compared to using the normal light stick?” There were 46 valid responses (40 males and 6 females, aged 16–57, mean age of 30.4). The average scores for the questionnaire are shown in Fig. 5 (right). As in the previous preliminary experiments, the audience reported that our proposed system contributed to a sense of unity when asked to compare it to the use of ordinary light sticks.

5 DISCUSSION

From the experiments and evaluation, it was shown that the transmission of a performer’s physical action via the vibration of a wearable device contributed to a sense of unity between the performer and the audience. The feedback from other people’s activities in a group experience is considered to have a secondary effect of strengthening the empathic experience among the audience, as well as making the presence of others (performers) felt strongly, as in some attempts that focus on the dialogue between individuals in previous research [2, 10, 11, 18]. The sharing of physical activities in a group experience provides a powerful interface for connecting with content and the influencers sharing it. It could also have useful value as a tool for creating co-experiences.

From the comparison of the unidirectional and bidirectional systems, it was found that the reflection of the audience’s cheering

behavior on the illuminated costumes contributed to the enhancement of the sense of unity among the audience. In the proposed system, the support focused on audience members’ common physical behavior, such as swinging cheering items. This accelerated the sharing of a sense of purpose in the group and promoted the unification of behavior. This kind of framework for supporting unification in groups is also applicable to supporting other cases where collective effervescence is observed, such as at sporting events. Moreover, collective co-actions involving physical activity, such as toasting and applauding at receptions, are events that make individuals perceive that they are there as part of a group. The framework of affording physical co-actions has the potential to be applied as a means of involving various individuals in a wide variety of group settings with diverse group membership.

In addition, a unique feature of our system is the use of cheering items as user devices. In the context of live entertainment, there have been several proposals involving the smartphones of audience members [28, 29], but directing the audience’s attention to a tool that is not part of the existing live performance scheme interferes with the experience value and concentration. In this system, a cheering item that was already included in interaction schemes to express one’s support was used as a device to provide feedback and assistance to the audience members and improve their experience and participation. Feedback to physical devices used specifically in group interaction and enhancing their functions can enhance the sense of immersion in the content and provide a means for active self-expression and recognition of the other person’s state in various group communications.

Although in this paper we have focused our investigation on the system’s contribution to collective effervescence in a local live performance environment, the proposed method can extend easily to support remote participants by transmitting over the Internet. The extension of the system addresses the scalability problem associated with the capacity of the communication system regarding a local venue and could facilitate the contribution of diverse people to the collective energy. This would help remove local and remote barriers with a schema that involves the same physical movement as that found in the local environment. The feedback to the physical user device and the mechanism of affording physical movements that are encapsulated in our system can be used to extend existing systems of interaction between participants in remote live streaming that are currently limited to interacting through comments. Furthermore, the incorporation of a framework that contributes to the realization of enthusiastic on-site interaction into the support of remote participants, where the presence of others is difficult to sense due to the constraints of the experience modalities, may

contribute to the promotion of a sense of participation more than in the local environment.

6 CONCLUSION

In this study, we focused on the field of live performance as a place where a large number of people gather and interact with each other and proposed a support system for co-experience and physical co-actions among participants to enhance enthusiastic interaction between performers and audiences during live performances. From experiments and an evaluation in the form of live performances, we found that the interactive communication supports of physical actions contributed to the mutual integration of the performer and the audience. In the future, we plan to study how such a framework affects the sense of participation, presence, and coexistence with other audiences for participants in remote environments.

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